

Chuanruo Ning

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EDUCATION

Cornell University

Ph.D. in Computer Science. Advisors: Wei-Chiu Ma, Kuan Fang

Ithaca, NY, United States

Aug. 2024 – Present

Peking University

Turing Class, Bachelor of Science in Computer Science

Beijing, China

Aug. 2020 – Jun. 2024

RESEARCH EXPERIENCE

Research Intern, NVIDIA

Advised by Dr. Xinghao Zhu

Santa Clara, CA, United States

Jun. 2026 – Present

- Learning Dexterous Hand Manipulation using World Action Models

Ph.D. Student, Cornell University

Advised by Prof. Kuan Fang, Prof. Wei-Chiu Ma

Ithaca, NY, United States

Sep. 2024 – Present

- Adapting Generalist Robotic Policy to Novel Tasks by Proxy Steering
- Open-world Manipulation by Equipping VLM with Interactive Digital Twins

Research Intern, Johns Hopkins University

Advised by Prof. Alan Yuille

Baltimore, MD, United States

Jun. 2023 – Jan. 2024

- 3D Part Recognition from 2D Image with a Single 3D Annotation

Research Assistant, Peking University

Advised by Prof. Hao Dong, Dr. Kaichun Mo, Dr. Ruihai Wu

Beijing, China

Jan. 2022 – Jul. 2023

- Few-shot Affordance Learning for Articulated Objects
- Foresightful Deformable Object Manipulation

PUBLICATIONS

* indicates equal contributions, † indicates equal advising

- [1] **Proxy Policy Steering** CoRL, 2026
Chuanruo Ning*, Tianrui Wang*, Wei-Chiu Ma†, Kuan Fang†
- [2] **X-Diffusion: Training Diffusion Policies on Cross-Embodiment Human Demonstrations** ICRA, 2025
Maximus Pace*, Prithwish Dan*, Chuanruo Ning, Atiksh Bhardwaj, Wei-Chiu Ma, Kushal Kedia
- [3] **Prompting with the Future: Open-World Model Predictive Control with Interactive Digital Twins** RSS, 2025
Chuanruo Ning, Kuan Fang†, Wei-Chiu Ma†
- [4] **Broadcasting Support Relations Recursively from Local Dynamics for Object Retrieval in Clutters** RSS, 2024
Yitong Li*, Ruihai Wu*, Haoran Lu, Chuanruo Ning, Yan Shen, Guanqi Zhan, Hao Dong
- [5] **Where2Explore: Few-shot Affordance Learning for Unseen Novel Categories of Articulated Objects** NeurIPS, 2023
Chuanruo Ning, Ruihai Wu, Haoran Lu, Kaichun Mo, Hao Dong
- [6] **Learning Environment-Aware Affordance for 3D Articulated Object Manipulation under Occlusion** NeurIPS, 2023
Kai Cheng*, Ruihai Wu*, Yan Shen, Chuanruo Ning, Guanqi Zhan, Hao Dong
- [7] **Learning Foresightful Dense Visual Affordance for Deformable Object Manipulation** ICCV, 2023
Ruihai Wu*, Chuanruo Ning*, Hao Dong

SELECTED HONORS AND AWARDS

- Huatai Securities Scholarship, 2024
- Merit Student, Peking University, 2023
- John Hopcroft Scholarship, Peking University, 2022
- Dean's Scholarship, Peking University, 2022

ACADEMIC SERVICE

- Conference Reviewer: NeurIPS, ICLR, CoRL, ICRA, CVPR, ICCV, ECCV, 3DV, AAAI